HUNT-SAGE · THREAT HUNTING CHEAT SHEET

Sysmon Event IDs for Hunters

Every Sysmon v15 event with its key fields, volume and the techniques it exposes, the field conventions that make pivots work, config fragments worth copying, and the SPL, KQL and Defender equivalents.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-11-03

Sysmon (Sysinternals, v15.x, schema 4.90) writes to `Microsoft-Windows-Sysmon/Operational`. Install with a configuration (`sysmon64 -accepteula -i config.xml`), update with `-c config.xml`, and start from a maintained template rather than a blank file; see Labs for a lab build. Event numbers are stable across versions; what changes is the schema and the filterable fields.

Events

ID	Event	Key fields	Volume	Techniques	Watch for
1	ProcessCreate	`Image`, `CommandLine`, `ParentImage`, `ParentCommandLine`, `User`, `ParentUser`, `IntegrityLevel`, `Hashes`, `OriginalFileName`, `CurrentDirectory`, `ProcessGuid`, `ParentProcessGuid`, `LogonGuid`	High	T1059, T1218, T1036, T1053	Rare parent/child pairs, `OriginalFileName` not matching `Image`, Medium parent with High child, interpreters spawned by Office or browsers
2	FileCreateTime	`TargetFilename`, `CreationUtcTime`, `PreviousCreationUtcTime`, `Image`	Low	T1070.006	Executables or scripts whose creation time is moved back; ignore browsers and installers
3	NetworkConnect	`Image`, `DestinationIp`, `DestinationHostname`, `DestinationPort`, `SourcePort`, `Initiated`, `Protocol`, `User`	Very high	T1071, T1021, T1048, T1095	Process-attributed beacons, `rundll32` / `mshta` / `regsvr32` with outbound connections, servers initiating to the internet
4	Sysmon service state changed	`State`, `Version`, `SchemaVersion`	Low	T1562.001	Any `Stopped` outside patching: tamper
5	ProcessTerminate	`Image`, `ProcessGuid`	High	T1059	Lifetime of short-lived tools when joined to event 1
6	DriverLoad	`ImageLoaded`, `Hashes`, `Signed`, `Signature`, `SignatureStatus`	Low	T1068, T1014, T1562.001	Unsigned or expired-cert drivers, known vulnerable drivers (BYOVD: loldrivers.io hashes), drivers from `C:\Users` or `C:\Windows\Temp`
7	ImageLoad	`Image`, `ImageLoaded`, `Signed`, `SignatureStatus`, `OriginalFileName`, `Hashes`	Very high (filter)	T1574.001/.002, T1055, T1003	Unsigned DLLs loaded from user-writable paths, system DLL names loaded from the wrong path, `samlib.dll` / `vaultcli.dll` / `wincard.dll` in odd processes, `clr.dll` in `rundll32` / `mshta`
8	CreateRemoteThread	`SourceImage`, `TargetImage`, `StartAddress`, `StartModule`, `StartFunction`	Low	T1055.001/.003	Source not in a known injector allow-list, `StartModule` empty, target is `lsass.exe`, `explorer.exe` or a browser
9	RawAccessRead	`Image`, `Device`	Low	T1003.003, T1006	Reads of `\Device\HarddiskVolume*` by anything but backup, AV or `vssvc.exe`: NTDS.dit and SAM theft

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ID	Event	Key fields	Volume	Techniques	Watch for
10	ProcessAccess	`SourceImage`, `TargetImage`, `GrantedAccess`, `CallTrace`, `SourceUser`, `TargetUser`	High (filter)	T1003.001, T1055	Access to `lsass.exe` with credential-dump masks (below), `CallTrace` with `UNKNOWN` frames, access from `C:\Users`
11	FileCreate	`TargetFilename`, `Image`, `CreationUtcTime`, `User`	High (filter)	T1105, T1074, T1547.001, T1486	Executables and scripts in `Temp`, `Public`, `ProgramData`, `Startup`; ransom notes; `.lnk` / `.iso` / `.img` from mail and browsers
12	RegistryEvent (object create / delete)	`EventType`, `TargetObject`, `Image`	High (filter)	T1547, T1546, T1112	New keys under `Run`, `Services`, `Winlogon`, `Image File Execution Options`, CLSID `InprocServer32`
13	RegistryEvent (value set)	`EventType`, `TargetObject`, `Details`, `Image`	High (filter)	T1547.001, T1546.015, T1562.001, T1112	`Details` pointing at user paths or script hosts, Defender `Exclusions`, `DisableAntiSpyware`, `EnableLUA`, `LocalAccountTokenFilterPolicy`, `ms-settings` shell handlers (UAC bypass)
14	RegistryEvent (key and value rename)	`EventType`, `TargetObject`, `NewName`	Low	T1112	Renames that hide persistence
15	FileCreateStreamHash	`TargetFilename`, `Contents`, `Hash`, `Image`	Medium	T1564.004, T1553.005	`Zone.Identifier` with `HostUrl` (download origin for free), payloads hidden in ADS
16	ServiceConfigurationChange (Sysmon config)	`Configuration`, `ConfigurationFileHash`	Low	T1562.001	Any config change not matching your deployment hash
17	PipeEvent (pipe created)	`PipeName`, `Image`, `User`	Medium	T1021.002, T1071, T1055	C2 pipe names (`\msagent_*`, `\MSSE-*-server`, `\postex_*`, `\mojo.*` lookalikes), `\PSEXESVC`, `\paexec*`, `\csexec*`, `\atsvc` / `\srvsvc` from non-system images
18	PipeEvent (pipe connected)	`PipeName`, `Image`	Medium	T1021.002, T1055	Clients of the pipes above; `\lsass` / `\samr` / `\winreg` connections from unexpected images
19	WmiEvent (filter activity)	`Operation`, `User`, `EventNamespace`, `Name`, `Query`	Low	T1546.003	`__EventFilter` creation with `Win32_ProcessStartTrace`, `__InstanceModificationEvent` timers
20	WmiEvent (consumer activity)	`Operation`, `Name`, `Type`, `Destination`	Low	T1546.003	`CommandLineEventConsumer` / `ActiveScriptEventConsumer`; `Destination` is the payload
21	WmiEvent (consumer to filter binding)	`Operation`, `Consumer`, `Filter`	Low	T1546.003	The binding is the persistence; alert on create
22	DNSEvent	`QueryName`, `QueryStatus`, `QueryResults`, `Image`, `User`	Very high (filter)	T1071.004, T1568, T1048.003	Process-attributed DNS: long or high-entropy labels, TXT-heavy clients, DGA-like NXDOMAIN bursts, `QueryResults` with `::ffff:` IPv4 mapping
23	FileDelete (archived)	`TargetFilename`, `Image`, `Hashes`, `IsExecutable`, `Archived`	Medium	T1070.004, T1485	Deletion of dropped tools; the file is copied to `ArchiveDirectory` for recovery
24	ClipboardChange	`Image`, `Session`, `ClientInfo`, `Hashes`	Low (opt-in)	T1115	RDP clipboard use; sensitive, enable with care
25	ProcessTampering	`Image`, `Type` (`Image is replaced`, `Image is locked for access`)	Low	T1055.012, T1036	Hollowing, herpaderping, ghosting; filter the known-good ones (browsers, `WerFault`, some installers)

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ID	Event	Key fields	Volume	Techniques	Watch for
26	FileDeleteDetected	`TargetFilename`, `Image`, `Hashes`, `IsExecutable`	Medium	T1070.004, T1486	Same as 23 without the archive copy; mass deletes by one process, shadow-copy-adjacent cleanup
27	FileBlockExecutable	`TargetFilename`, `Image`, `Hashes`	Low	T1105, T1204	Blocked PE writes (for example Office or browsers writing `.exe`); the block is the detection
28	FileBlockShredding	`TargetFilename`, `Image`, `IsExecutable`	Low	T1485	Blocked overwrite-delete patterns used by shredders and wipers
29	FileExecutableDetected	`TargetFilename`, `Image`, `Hashes`, `User`	Medium	T1105, T1204.002	New PE files written anywhere: cheap first-seen inventory of binaries per host
255	Error	`ID`, `Description`	Low		Driver or service faults; a burst usually means a config problem or tampering

Field conventions that make pivots work

- **`ProcessGuid` / `ParentProcessGuid`**: unique per process instance across reboots; build trees on these, never on PID.
- **`LogonGuid` and `LogonId`**: join to Security 4624 (`TargetLogonId`) to get logon type and source for any process.
- **`OriginalFileName`** comes from the PE version resource and survives renaming; compare with the `Image` basename (`rclone.exe` renamed to `svchost.exe`, `procdump` renamed to `dump.exe`).
- **`Hashes`**: configure `<HashAlgorithms>md5,sha256,IMPHASH</HashAlgorithms>`; IMPHASH clusters recompiled tooling, SHA256 is for prevalence (count of hosts per hash).
- **`User` / `ParentUser` / `SourceUser` / `TargetUser`** (v13+): catches cross-user injection and services spawning user-context children.
- **`IntegrityLevel`**: `Low` (browser sandboxes), `Medium` (users), `High` (elevated), `System`; a High child of a Medium parent without a consent prompt is a UAC bypass lead.
- **`CallTrace`** (event 10): module-by-module stack of the caller; `UNKNOWN(...)` frames mean memory not backed by a file (injected or reflective code), `dbgcore.dll` / `dbghelp.dll` means a minidump API.
- **`GrantedAccess`** (event 10) masks seen against `lsass.exe`: `0x1010` (PROCESS_VM_READ + QUERY_LIMITED_INFORMATION: Mimikatz-style read), `0x1410`, `0x1438`, `0x143A`, `0x1FFFFF` (PROCESS_ALL_ACCESS: procdump, comsvcs MiniDump), `0x0010` (VM_READ only), `0x0040` (DUP_HANDLE), `0x1F3FFF`; `0x1000` or `0x1400` alone (query only) is usually benign.
- **`RuleName`**: tag every include rule with the technique it serves (`RuleName="technique_id=T1003.001,technique_name=LSASS Memory"`); the tag arrives in the event and makes triage and coverage counting trivial.
- **`QueryStatus`** (event 22): `0` success, `9003` NXDOMAIN (NAME_ERROR), `9501` no records, `1460` timeout; NXDOMAIN bursts per process are the DGA signal.
- **`Zone.Identifier`** (event 15): `Contents` holds `ZoneId=3` plus `HostUrl` / `ReferrerUrl`; the origin of every downloaded file without a proxy log.

High-value filters (fragments to copy)

Schema 4.90; `onmatch="include"` logs only what matches, `onmatch="exclude"` logs everything except. For event 1 log nearly everything and exclude noise; for 3, 7, 11, 12/13 and 22 use includes.

```
<!-- 10: anything touching LSASS with a read or full-access mask -->
<ProcessAccess onmatch="include">
  <Rule groupRelation="and" name="technique_id=T1003.001,technique_name=LSASS Memory">
    <TargetImage condition="is">C:\Windows\system32\lsass.exe</TargetImage>
    <GrantedAccess condition="contains any">0x1010;0x1410;0x1438;0x143A;0x1FFFFF;0x0010;0x1F3FFF</GrantedAccess>
  </Rule>
</ProcessAccess>
<!-- 7: unsigned DLLs from user-writable paths, and classic side-load targets -->
<ImageLoad onmatch="include">
  <Rule groupRelation="and" name="technique_id=T1574.002,technique_name=DLL Side-Loading">
    <ImageLoaded condition="begin with">C:\Users\</ImageLoaded>
    <Signed condition="is">false</Signed>
  </Rule>
  <ImageLoaded condition="end with" name="technique_id=T1574.001">version.dll;wininet.dll;cryptbase.dll;dbghelp.dll;
mscoree.dll;userenv.dll;profapi.dll;dwmapi.dll;uxtheme.dll;wtsapi32.dll</ImageLoaded>
  <ImageLoaded condition="end with" name="technique_id=T1003">samlib.dll;vaultcli.dll;winscard.dll;ntdsapi.dll;comsv
cs.dll</ImageLoaded>
</ImageLoad>
<!-- 17/18: C2 and remote-execution pipes -->
<PipeEvent onmatch="include">
  <PipeName condition="contains any" name="technique_id=T1071,technique_name=C2 named pipes">\msagent_;\MSSE-;\poste
x_;\status_;\mypipe;\win_svc;\ntsvcs_;\scerpc_;\DserNamePipe;\SearchTextHarvester;\wkssvc_;\spoolss_;\winsock</PipeN
ame>
  <PipeName condition="contains any" name="technique_id=T1021.002,technique_name=Remote service execution">\PSEXESV
C;\paexec;\csexec;\remcom;\atsvc</PipeName>
</PipeEvent>
<!-- 11: executables and scripts landing in writable and autostart locations -->
<FileCreate onmatch="include">
  <TargetFilename condition="contains any" name="technique_id=T1547.001,technique_name=Startup Folder">\Start Menu\P
rograms\Startup\;\Startup\</TargetFilename>
  <Rule groupRelation="and" name="technique_id=T1105,technique_name=Ingress Tool Transfer">
    <TargetFilename condition="contains any">\AppData\Local\Temp\;\Users\Public\;C:\ProgramData\;C:\Windows\Temp\;\D
ownloads\</TargetFilename>
    <TargetFilename condition="end with">.exe;.dll;.ps1;.vbs;.js;.hta;.bat;.cmd;.scr;.iso;.img;.lnk;.msi;.cpl</Targe
tFilename>
  </Rule>
</FileCreate>
<!-- 12/13: persistence and tamper keys -->
<RegistryEvent onmatch="include">
  <TargetObject condition="contains any" name="technique_id=T1547.001,technique_name=Registry Run Keys">\CurrentVers
ion\Run;\CurrentVersion\RunOnce;\CurrentVersion\Explorer\Shell Folders;\CurrentVersion\Winlogon\Shell;\CurrentVersio
n\Winlogon\Userinit;\CurrentVersion\Winlogon\Notify</TargetObject>
  <TargetObject condition="contains any" name="technique_id=T1543.003,technique_name=Windows Service">\Services\;\Im
agePath;\ServiceDll</TargetObject>
  <TargetObject condition="contains any" name="technique_id=T1546,technique_name=Event Triggered Execution">\Image F
ile Execution Options\;\InprocServer32;\LocalServer32;\ScriptletURL;\TreatAs;\AppCertDlls;\AppInit_DLLs;\SilentProce
ssExit\;\ms-settings\Shell\Open\command;\mscfile\shell\open\command</TargetObject>
  <TargetObject condition="contains any" name="technique_id=T1562.001,technique_name=Impair Defenses">\Windows Defen
der\Exclusions\;\Windows Defender\DisableAntiSpyware;\Real-Time Protection\;\Policies\System\EnableLUA;\LocalAccount
TokenFilterPolicy;\Lsa\RunAsPPL;\DisableRestrictedAdmin;\Terminal Server\fDenyTSConnections</TargetObject>
</RegistryEvent>
<!-- 22: cut the noise, keep the signal -->
<DnsQuery onmatch="exclude">
  <QueryName condition="end with">.microsoft.com;.windowsupdate.com;.office.net;.in-addr.arpa;.local</QueryName>
  <Image condition="is">C:\Windows\System32\svchost.exe</Image>
</DnsQuery>
<!-- 27: block (and log) Office and browsers writing executables -->
<FileBlockExecutable onmatch="include">
  <Image condition="end with" name="technique_id=T1204.002,technique_name=Malicious File">\winword.exe;\excel.exe;\p
owerpnt.exe;\outlook.exe;\msedge.exe;\chrome.exe;\firefox.exe</Image>
</FileBlockExecutable>
```

Configuration tips

- **Templates**: SwiftOnSecurity `sysmon-config` (one file, conservative, readable), Olaf Hartong `sysmon-modular` (per-technique modules merged by a script, `RuleName` tags included, heavier), Florian Roth's `sysmon-config` fork (adds more evasion coverage). Start with one, diff against the others, keep your own in version control.
- **Include vs exclude**: exclude-based logging for 1 (see everything, trim known-noisy parents such as updaters) and 5; include-based for 3, 7, 10, 11, 12, 13, 17, 18, 22. Mixing both for one event needs two rule groups; order and `groupRelation` matter.
- **Noise to expect**: 7 and 22 dominate volume; 25 fires for browsers, `WerFault.exe` and some installers; 6 fires at boot for every driver, so filter by `Signed` / `SignatureStatus` rather than dropping it; 3 from `svchost`, Teams and browsers.
- **Event 23 archiving**: `<ArchiveDirectory>Sysmon</ArchiveDirectory>` keeps deleted files under `C:\Sysmon\` (system ACL); set a size limit and a cleanup job, or use 26 instead if storage is tight.
- **Tamper detection**: alert on event 4 (`Stopped`), 16 (config hash change), System 7036 for `Sysmon64` stopping, and on `sysmon64.exe -u` / `-c` in event 1. Rename the service and driver (`-d` at install) to raise the bar.

- **Deployment**: GPO startup script or Intune Win32 app; push config updates with `sysmon64 -c new.xml`; keep the config hash in a registry value or in `RuleName` so you can tell which hosts run which version. Dump the live config with `sysmon64 -c` (no file).
- **Sysmon for Linux** (`sysmonforlinux`): events 1, 3, 5, 9, 10, 11, 16, 23 only; writes XML to syslog (`/var/log/syslog` or journald, facility `local0`), same schema names, no hashes of images by default; useful for `execve` and network attribution where auditd is too raw.
- **Collect it**: WEF subscription or agent on `Microsoft-Windows-Sysmon/Operational`; keep XML rendering so field names survive.

Query examples

Splunk (sourcetype `XmlWinEventLog:Microsoft-Windows-Sysmon/Operational`, or `sysmon` via the Sysmon add-on; field names match the event fields):

```
index=sysmon EventCode=1 | eval parent=lower(ParentImage), child=lower(Image) | stats dc(host) as hosts count values
(CommandLine) as cmd by parent child | where hosts<=2
index=sysmon EventCode=1 | eval img=lower(replace(Image,".*\\\\","")) | where OriginalFileName!="" AND lower(Origina
lFileName)!=img AND lower(OriginalFileName)!=img+".exe" | stats count by host Image OriginalFileName
index=sysmon EventCode=3 Initiated=true | stats count dc(DestinationIp) as dests by host Image | where dests>50 AND
NOT match(Image,"(?i)chrome|msedge|firefox|teams|svchost")
index=sysmon EventCode=7 Signed=false ImageLoaded="C:\\Users\\*" | stats dc(host) as hosts by ImageLoaded Image | so
rt hosts
index=sysmon EventCode=10 TargetImage="*\\lsass.exe" GrantedAccess IN ("0x1010","0x1410","0x1438","0x143A","0x1FFFF
F") | stats count by host SourceImage GrantedAccess CallTrace
index=sysmon EventCode=11 TargetFilename IN ("*\\Temp\\*.exe","*\\Public\\*.exe","*\\ProgramData\\*.exe","*\\Startup
\\*") | stats count by host Image TargetFilename
index=sysmon EventCode=13 TargetObject="*\\CurrentVersion\\Run*" | stats dc(host) as hosts by TargetObject Details |
where hosts<=3
index=sysmon EventCode IN (17,18) | stats count dc(host) as hosts by PipeName Image | sort hosts count
index=sysmon EventCode=22 QueryStatus=9003 | stats dc(QueryName) as nx by host Image | where nx>100
index=sysmon EventCode=25 | stats count by host Image Type | where NOT match(Image,"(?i)msedge|chrome|firefox|werfau
lt")
```

Microsoft Sentinel (Sysmon forwarded into the `Event` table; the Sysmon parser or ASIM `imProcessCreate` gives typed columns, otherwise extract from `EventData`):

```
Event
| where Source == "Microsoft-Windows-Sysmon" and EventID == 1
| extend d = parse_xml(EventData).DataItem.EventData.Data
| mv-apply d on (summarize b = make_bag(pack(tostring(d["@Name"]), tostring(d["#text"]))))
| extend Image = tostring(b.Image), Parent = tostring(b.ParentImage), Cmd = tostring(b.CommandLine), Orig = tostring
(b.OriginalFileName)
| where isnotempty(Orig) and tolower(Orig) != tolower(split(Image, "\\")[-1])
| summarize count(), any(Cmd) by Computer, Image, Orig
```

Defender for Endpoint equivalents when you have EDR instead of (or as well as) Sysmon:

Sysmon	Defender for Endpoint table	Notes
1 / 5	`DeviceProcessEvents`	`ProcessVersionInfoOriginalFileName` = `OriginalFileName`, `ProcessIntegrityLevel`, `InitiatingProcess*` = parent
3	`DeviceNetworkEvents`	`ActionType` `ConnectionSuccess` / `ConnectionFailed` / `InboundConnectionAccepted`; `RemoteUrl` when known
6 / 7	`DeviceImageLoadEvents`	Drivers and DLLs together; `SHA256`, `FileName`, `FolderPath`
8 / 10 / 25	`DeviceEvents`	`ActionType` in (`CreateRemoteThreadApiCall`, `OpenProcessApiCall`, `ReadProcessMemoryApiCall`, `NtAllocateVirtualMemoryRemoteApiCall`, `NtMapViewOfSectionRemoteApiCall`, `SetThreadContextRemoteApiCall`, `QueueUserApcRemoteApiCall`); LSASS access is `OpenProcessApiCall` with `FileName == "lsass.exe"` and the mask in `AdditionalFields.DesiredAccess`
11 / 23 / 26 / 29	`DeviceFileEvents`	`ActionType` `FileCreated`, `FileModified`, `FileDeleted`, `FileRenamed`; `FileOriginUrl` / `FileOriginReferrerUrl` replaces the Zone.Identifier trick
12 / 13 / 14	`DeviceRegistryEvents`	`ActionType` `RegistryKeyCreated`, `RegistryValueSet`, `RegistryKeyDeleted`; `RegistryValueData` = `Details`
15	`DeviceFileEvents`	`FileOriginUrl`
17 / 18	`DeviceEvents`	`ActionType` `NamedPipeEvent`, pipe name in `AdditionalFields`

Sysmon	Defender for Endpoint table	Notes
19 / 20 / 21	`DeviceEvents`	`ActionType` `WmiBindEventFilterToConsumer` and related
22	`DeviceNetworkEvents` / `DeviceEvents`	`ActionType` `DnsQueryResponse` (`DeviceEvents`, `AdditionalFields` carries the query and response)

Related: Windows Security event IDs, LOLBins, playbooks for LSASS credential dumping, DLL hijacking, Process injection, Persistence and Defense evasion; Emulate and detect to prove each filter fires.

HUNT-SAGE
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